

SESSION 3

RAD-201 Basic Physics and Electronics

X-ray Circuitry, Transformers and Generator Control

How primary and secondary circuits create filament current, tube voltage and controlled X-ray production

Course	RAD-201 Radiation Physics
Date	2026
Lecturer	A. A. Zubair
Suggested study time	240–300 minutes

Learning Objectives

By the end of this session, you should be able to:

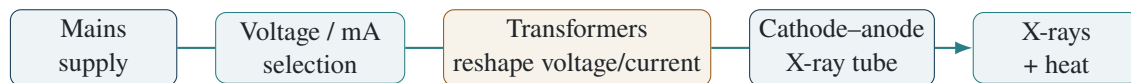
1. explain why the X-ray tube requires both a low-voltage/high-current filament supply and a high-voltage tube supply;
2. identify the primary and secondary windings of a transformer and explain electromagnetic induction in simple terms;
3. use the transformer turns-ratio equations to distinguish step-up and step-down transformers;
4. explain the roles of the autotransformer, filament transformer and high-tension transformer in an X-ray generator;
5. relate kVp, mA, exposure time and mAs to the appropriate parts of the X-ray circuit;
6. explain rectification, voltage ripple and the advantage of high-frequency generators; and
7. use circuit knowledge to interpret simple generator faults and exposure problems safely.

The circuit has two electrical jobs

An X-ray tube needs **many electrons** and it needs those electrons to have **high kinetic energy**. The filament circuit provides **low voltage and high current** to heat the cathode and release electrons. The high-voltage circuit provides a **large potential difference (kVp)** to accelerate those electrons toward the anode. Transformers allow the generator to create both conditions from the available electrical supply.

1 Session Orientation: follow the electron

The easiest way to understand the X-ray generator is to follow one electron from the electrical supply to the tungsten target.



Three questions organize the whole session:

1. **How are electrons released?** The filament is heated by a low-voltage, high-current supply.
2. **How are electrons accelerated?** A step-up transformer creates the high tube potential selected as kVp.
3. **How is the exposure controlled?** Rectifiers, timers, electronic switching and AEC control direction, duration and consistency.

Radiography link

When you select kVp and mAs at the console, you are not simply choosing numbers. You are asking the generator to establish specific electrical conditions in different parts of the circuit before and during the exposure.

2 Why transformers are central to X-ray production

The electrical supply entering an X-ray room is not suitable for direct connection to the X-ray tube. The cathode filament needs only a few volts but several amperes of heating current. In contrast, useful diagnostic X-ray production requires tens to more than one hundred kilovolts across the tube.

Transformer Definition

A **transformer** is an electrical device that transfers electrical energy between circuits by **electromagnetic induction**. It changes voltage and current according to the turns ratio between its **primary** and **secondary** windings while keeping the circuits magnetically coupled through a core.

2.1 Primary and secondary windings

A basic transformer contains two windings linked by a magnetic core. The **primary winding** receives the input alternating voltage. The **secondary winding** supplies the transformed output to the next part of the circuit. Alternating current in the primary produces a changing magnetic field in the core; that changing magnetic flux induces a voltage in the secondary. The windings therefore transfer energy magnetically rather than by direct electrical contact.

A transformer needs changing current

Ordinary transformer action requires a changing magnetic field, so transformers operate with AC or with rapidly switched current that behaves like AC. A steady DC current does not continuously induce a secondary voltage after the initial change.

3 The transformer ratios

For an ideal transformer,

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}.$$

If the secondary has more turns than the primary, voltage is stepped up. If it has fewer turns, voltage is stepped down.

Current changes in the opposite direction. For an ideal transformer,

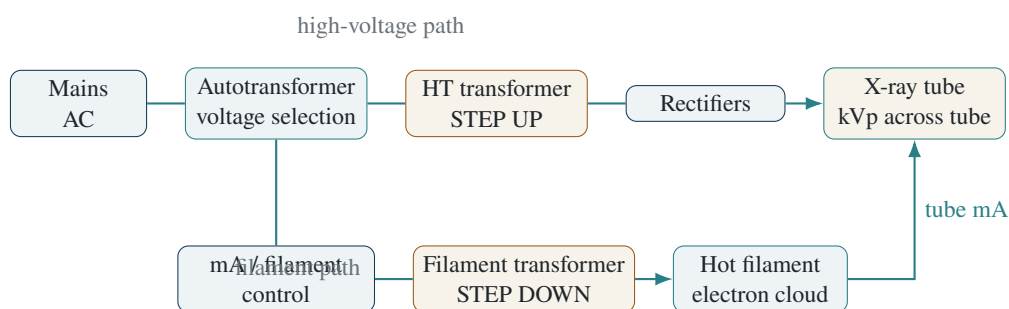
$$V_P I_P \approx V_S I_S, \quad \frac{I_S}{I_P} = \frac{N_P}{N_S}.$$

Thus a voltage step-up is accompanied by a current step-down, while a voltage step-down is accompanied by a current step-up. Real transformers have losses, so output power is slightly less than input power.

Transformer	Turns / voltage relationship	Radiography meaning
Step-up	$N_S > N_P$, so $V_S > V_P$	Used in the high-tension circuit to produce kV across the X-ray tube
Step-down	$N_S < N_P$, so $V_S < V_P$	Used in the filament circuit to obtain low voltage and high heating current
Autotransformer	One tapped winding; selected turns vary output	Used for voltage selection/control before the high-voltage stage

4 The complete X-ray circuit: two paths working together

The X-ray generator can be simplified into a **filament path** and a **high-voltage path**. Both begin from the supply/control system and meet functionally at the X-ray tube.



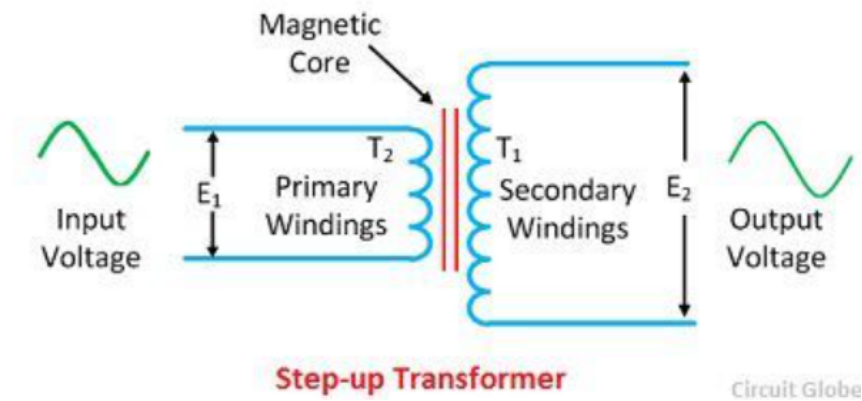
Do not mix up the two transformers

High-tension transformer: step-up voltage → creates the selected kVp across the tube.

Filament transformer: step-down voltage and step-up current → heats the filament → controls electron supply and therefore tube current (mA).

5 The high-tension transformer: producing kVp

The **high-tension (HT) transformer** has many more turns on its secondary than on its primary. It therefore raises the generator voltage into the diagnostic kilovoltage range.



Step-up transformer: the secondary has more turns than the primary, so the output voltage is increased. In the X-ray generator this principle is used to create the high tube potential (kVp).

The secondary output is connected, through rectification and high-voltage circuitry, across the cathode and anode. The resulting potential difference accelerates electrons across the evacuated tube. The maximum kinetic energy gained by one electron is approximately equal to the selected tube potential expressed in keV:

$$100 \text{ kVp} \Rightarrow \text{electron kinetic energy up to about } 100 \text{ keV}.$$

Step-up transformer turns ratio

A high-tension transformer receives 500 V at its primary and must provide 100 kV at the secondary. The required turns ratio is

$$\frac{N_S}{N_P} = \frac{V_S}{V_P} = \frac{100000}{500} = 200.$$

So the secondary needs about **200 times as many turns** as the primary. If the primary had 600 turns, an idealized secondary would require

$$N_S = 200 \times 600 = 120000 \text{ turns}.$$

The numerical values are simplified for teaching, but the principle is exactly the one used for voltage step-up.

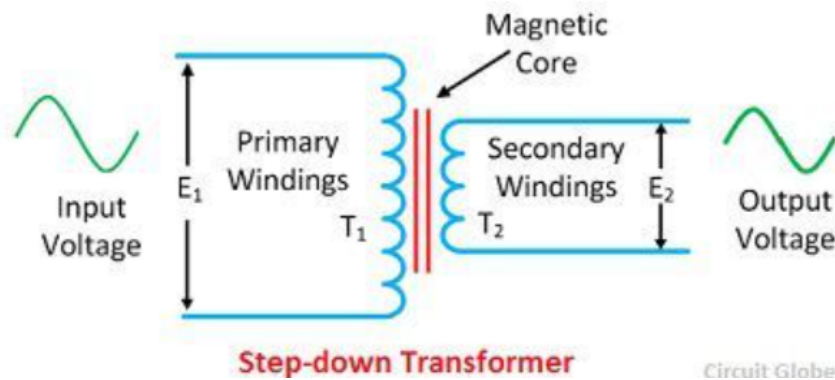
Why this matters clinically

When kVp is increased, the generator establishes a larger potential difference across the X-ray tube. Electrons reach the target with greater maximum kinetic energy, producing a more penetrating beam and generally increasing X-ray output.

6 The filament transformer: low voltage, high heating current

The filament is a small tungsten coil. It must become very hot to produce **thermionic emission**, but applying kilovoltage directly to the filament heater would be inappropriate. The filament transformer therefore performs the opposite job to the HT transformer.

Its secondary has fewer turns than its primary, so the secondary voltage is low. Because transformer power is approximately conserved, the available secondary current becomes much higher. This large current heats the filament by resistive heating.



Step-down transformer: the secondary has fewer turns than the primary, so voltage falls while available current rises. In the filament circuit this provides low voltage and high current for heating the tungsten filament.

Step-down filament transformer

A filament transformer has a 240 V primary and a 12 V secondary. Its voltage ratio is

$$\frac{V_S}{V_P} = \frac{12}{240} = \frac{1}{20}.$$

If the ideal primary current is 0.25 A, approximate power conservation gives

$$V_P I_P \approx V_S I_S,$$

$$(240)(0.25) = (12)I_S, \quad I_S \approx 5 \text{ A}.$$

The transformer has therefore changed a relatively high-voltage, low-current input into a **low-voltage, high-current output suitable for filament heating**.

The filament transformer does not directly “step up tube mA”

Its high secondary current flows through the **filament heater**. That heating changes thermionic emission. The number of released electrons then determines how much **tube current** can flow from cathode to anode during the exposure. Filament current is measured in amperes; tube current is usually measured in milliamperes.

7 The autotransformer: where kVp selection begins

An **autotransformer** uses one tapped winding rather than separate primary and secondary windings. By selecting a different number of turns, the generator changes the voltage sent to the input of the high-voltage stage.

It is therefore associated with **kVp selection**, but it does not itself create the final kilovoltage across the X-ray tube. Think of it as the control stage that determines how much voltage the HT transformer will receive.

Autotransformer + HT transformer

Suppose the HT transformer has a step-up ratio of 400:1 and the desired tube potential is 80 kV. The primary voltage required by the HT transformer is

$$V_P = \frac{V_S}{400} = \frac{80000}{400} = 200 \text{ V}.$$

The autotransformer/control system therefore needs to supply about **200 V** to that primary in this simplified

example to obtain about **80 kV** at the secondary.

8 kVp and mA: two controls, two different physical effects

Control	Main circuit link	Main radiographic meaning
kVp	Autotransformer/control + HT step-up transformer	Maximum electron energy; beam quality/penetration; also influences output
mA	Filament control + filament step-down transformer	Number of electrons crossing the tube each second; strongly affects X-ray quantity
Time	Timer/electronic switching	Duration of electron flow and X-ray production
mAs	$\text{mA} \times \text{time}$	Total electron flow during the exposure; strongly related to photon quantity

mAs calculation

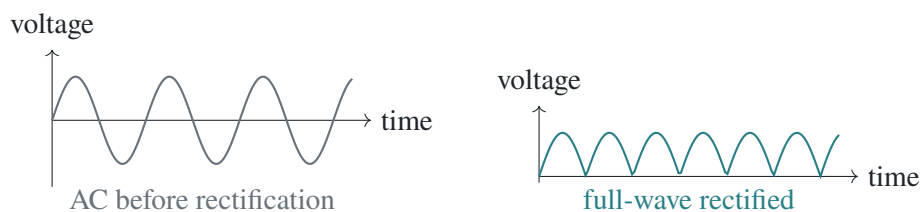
For an exposure of 320 mA for 0.025 s,

$$\text{mAs} = 320 \times 0.025 = 8 \text{ mAs.}$$

The same 8 mAs could be produced by 400 mA for 0.020 s. The shorter time can reduce motion blur, provided the tube/generator can safely supply the higher current.

9 Rectification: keeping electrons moving cathode to anode

Transformer output is alternating. The X-ray tube, however, requires the cathode to remain negative relative to the anode during useful electron acceleration. **Rectifiers** act as one-way electrical valves so that the high-voltage waveform applied to the tube has the correct polarity.



Full-wave rectification uses both halves of the AC cycle by making both produce the correct tube polarity. Modern systems use solid-state diodes and electronic power switching rather than older valve rectifiers.

10 Preparation, exposure timing and rotating-anode readiness

Before the actual X-ray exposure, a preparation stage may heat the selected filament and bring the rotating anode to operating speed. When the exposure condition is activated, the generator applies the selected high voltage for the required duration.

11 Voltage ripple and high-frequency generators

After rectification, tube voltage may still vary during the exposure. **Voltage ripple** describes the percentage variation between the maximum and minimum tube voltage. Larger ripple means electron energy changes more during each cycle; smaller ripple means the tube potential stays closer to the selected kVp.

High-frequency generators first convert the supply into rapidly switched electrical pulses. These high-frequency pulses can be transformed efficiently, rectified and smoothed to provide a compact system with low voltage ripple.

Why high frequency matters

For the same nominal kVp and mAs, a low-ripple high-frequency waveform maintains electron energy more consistently than an older high-ripple waveform. This generally produces more efficient and reproducible X-ray output.

Technique charts are equipment-specific

Two rooms set to the same displayed kVp and mAs can still give somewhat different output because generator waveform, filtration, tube condition and calibration differ. This is one reason exposure charts and QC measurements are tied to the specific X-ray unit.

12 Automatic exposure control fits into the timing system

Automatic exposure control (AEC) does not replace the X-ray generator. It provides an exposure-termination signal when the detector has received the predetermined amount of radiation.

The radiographer must still select an appropriate kVp, choose the correct chamber(s), centre the anatomy correctly and use suitable backup limits. AEC cannot correct poor positioning or an inappropriate chamber selection.

AEC does not control all exposure factors automatically

AEC mainly decides **when to terminate the exposure**. Generator design and the radiographer's selections still determine kVp, tube loading, chamber selection and many other conditions.

13 Using circuit knowledge to understand faults

A radiographer is not expected to repair the generator, but a logical circuit model helps explain abnormal behaviour.

Problem	Electrical effect	Possible radiographic consequence
Filament circuit failure	Inadequate heating / electron supply	Little or no tube current; exposure may fail
Filament regulation fault	Incorrect thermionic emission	Unexpected mA and output inconsistency
HT transformer/control fault	Selected tube potential not achieved	Incorrect penetration/output or exposure inhibition
Rectifier fault	Abnormal polarity/waveform	Reduced or unstable X-ray production
Timer/AEC fault	Exposure not terminated correctly	Underexposure or excessive patient/receptor exposure
Anode rotation failure	Poor heat distribution at target	Exposure may be inhibited to protect the tube

Danger Warning

You are a Radiographer, not an Engineer.

**Never touch the mains.
Always observe and report.**

Oversabi go kill u.

Circuit knowledge is for safe operation, recognition of abnormal equipment behaviour and accurate reporting. Internal mains and high-voltage generator components must be handled only by appropriately trained and authorized engineering/service personnel.

14 A self-explanatory mental model

If the circuit diagram seems complicated, reduce it to five questions:

1. **Where do the electrons come from?** Heated cathode filament.
2. **What controls how many are available?** Filament heating and mA regulation.
3. **What gives them kinetic energy?** High tube potential created by the step-up HT transformer.
4. **What keeps them moving in the correct direction?** Rectification.
5. **What decides how long X-rays are produced?** Timer, electronic switching or AEC.

Key Takeaways

- A transformer transfers AC electrical energy between windings by electromagnetic induction.
- The voltage ratio follows $V_S/V_P = N_S/N_P$; current changes inversely in an ideal transformer.
- The HT transformer is a step-up transformer that creates the kilovoltage range applied across the X-ray tube.
- The filament transformer is a step-down transformer that supplies low voltage and high heating current to the cathode filament.
- Filament current and tube current are related but are not the same current: filament heating controls

thermionic emission, which then controls tube mA.

- The autotransformer/control stage helps select the voltage supplied to the HT transformer and therefore participates in kVp selection.
- Rectification maintains the required cathode-to-anode electron flow; timing/AEC controls exposure duration.
- High-frequency generators reduce voltage ripple and provide more consistent tube potential and output.

Key Terms

Primary winding

Transformer winding connected to the input supply.

Secondary winding

Transformer winding from which transformed output is obtained.

Turns ratio

Ratio N_S/N_P that determines ideal transformer voltage ratio.

Step-up transformer

Transformer with higher secondary voltage; used in the X-ray high-tension circuit.

Step-down transformer

Transformer with lower secondary voltage and higher secondary current; used for filament heating.

Autotransformer

Single tapped winding used for variable voltage selection/control.

Rectification

Conversion of alternating voltage/current into a unidirectional form suitable for the tube.

Voltage ripple

Variation in tube potential during the exposure waveform.

mA

Tube current; the rate of electron flow from cathode to anode.

mAs

Product of tube current and exposure time; related to total electron flow during an exposure.

Self-Assessment Questions

1. Why does an X-ray generator require both a step-up transformer and a step-down transformer?
2. Define primary and secondary windings and explain electromagnetic induction in simple terms.
3. Write the ideal transformer voltage-ratio equation and state what makes a transformer step up or step down voltage.
4. A transformer receives 400 V and has a 250:1 secondary-to-primary turns ratio. What ideal secondary

voltage would be produced?

5. A filament transformer reduces 240 V to 12 V. What is the voltage turns ratio, and why does the secondary current increase?
6. Explain the difference between filament current and tube current (mA).
7. Describe how the autotransformer and HT transformer work together to determine kVp.
8. Why is rectification necessary after the high-voltage transformer?
9. Calculate the mAs for 500 mA at 0.016 s.
10. Explain why a high-frequency generator produces a more constant tube potential than an older high-ripple generator.
11. Give one likely radiographic consequence of a filament-circuit fault and one of an HT-circuit fault.

Series navigation

Next session: subsequent RAD-201 topics can build from this circuit foundation into X-ray production, generator waveforms, tube loading and other equipment physics. Keep the two-path model in mind: **filament circuit controls electron supply; high-voltage circuit controls electron energy.**

